they combine to a system of even parity. (ii) one particle in a state of even parity and one in a state of odd parity combine to a system of odd parity.

6.4. DEFORMED NUCLEI

6.4.1. Deformation index

Both the liquid-drop model and the single-particle model assume that the mass and charge of the nucleus are spherically symmetric. This is true only for nuclei close to the magic numbers; other nuclei have distorted shapes. The most common assumption about the distortion of the nuclide shape is that it is ellipsoidal, i.e. a latitudinal cross-section of the nucleus is an ellipse. Figure 6.6 shows the oblate (flying-saucer-like) and prolate (egg-shaped) ellipsoidally distorted nuclei; the prolate shape is the more common. Deviation from the spherical shape is given by

$$\beta = 2(a - c)/(a + c) \tag{6.38}$$

where a and c are the elliptical axes as shown on in Figure 6.6(c). For prolate shape $\beta > 0$, and for oblate shape $\beta < 0$. The maximum deformation observed is about $\beta = \pm 0.6$.

The deformation is related to the nuclear shell structure. Nuclei with magic numbers are spherical and have sharp boundary surfaces (they are "hard"). As the values of N and Z depart from the magic numbers the nucleus increases its deformation.

6.4.2. Electric multipoles

In a spherical nucleus we assume the charge distribution to be spherical and the nucleus acts as a monopole. In the deformed nuclei, the nuclear charge has a non-spherical distribution. The potential at a point, $\bar{x}$, $\bar{y}$, $\bar{z}$ (Fig. 6.6) will be found to vary depending on the charge distribution and mode of rotation of the nucleus. The nuclear charge may be distributed to form a dipole, a quadrupole, etc. Nuclei are therefore divided into

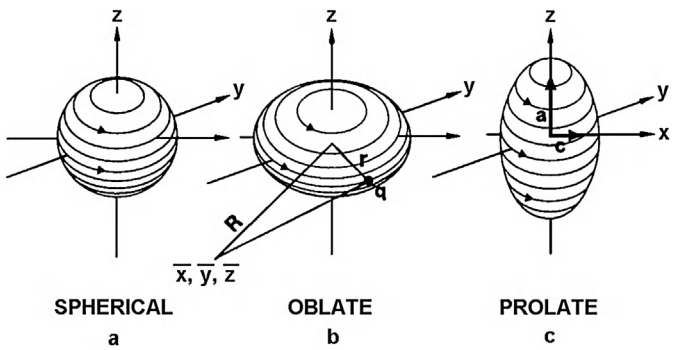


Figure 6.6 Three nuclidic shapes: (a) spherical nucleus, (b) oblate (extended at the equator), and (c) prolate (extended at the poles).

different classes depending on their electrical moments: monopoles, dipoles, quadrupoles, octupoles etc.

It has been found that nuclei with spin $I = 0$ have no multipole moment which corresponds to the reasoning made above regarding spherical shape and magic numbers. According to theory, nuclei with $I = ½$ can have a dipole moment, but this has not yet been shown experimentally. Nuclei with $I = 1$ have quadrupole moments; they are fairly common. The quadrupole moment $\hat{Q}$ can be calculated for spheroidal (i.e. deformation not too far from a sphere) nuclei in terms of the electron charge, **e**, by

$$\hat{Q} = (2/5)Z\left(a^2 - c^2\right) \tag{6.39}$$

$\hat{Q}$ is usually referred to as the internal quadrupole moment, i.e. the expected value for a rotation around the z-axis. However, quantum mechanics makes this impossible and gives for the maximum observable quadrupole moment

$$\hat{Q}_{\text{obs}} = \hat{Q}(I - 1/2)/(I + 1) \tag{6.40}$$

Thus $\hat{Q}_{\text{obs}} = 0$ for $I \leq 1/2$. $\hat{Q}_{\text{obs}}$ is usually given in area (m^2). Most commonly $10^{-28} m^2$ is used as unit and referred to as one barn, $\hat{Q}_{\text{obs}}$ is > 0 for the more common prolate shape, and < 0 for oblate. Some measured values are given in Table 6.3.

The rotation of nuclei with electric multipoles gives rise to formation of magnetic multipoles. Nuclei can therefore also be divided according to the magnetic moments in the same way as according to their electrical moments.

6.4.3. The collective nuclear model

In 1953 A. Bohr and Mottelsen suggested that the nucleus be regarded as a highly compressed liquid, undergoing quantized rotations and vibrations. Four discrete collective motions can be visualized. In Figure 6.6 we can imagine that the nucleus rotates around the y-axis as well as around the z-axis. In addition it may oscillate between prolate to oblate forms (so-called *irrotation*) as well as *vibrate*, for example, along the x-axis. Each mode of such collective nuclear movement has its own quantized energy. In addition, the movements may be coupled (cf. coupling of vibration and rotation in a molecule).

The model allows the calculation of rotational and vibrational levels as shown in Figure 6.7. If a ^{238}U nucleus is excited above its ground state through interaction with a high energy heavy ion (coulomb excitation), we have to distinguish between three types of excitation: (a) nuclear excitation, in which the quantum number j is changed to raise the nucleus to a higher energy level (according to Table 6.6); (b) vibrational excitation, in which case j is unchanged, but the nucleus is raised to a higher vibrational level, characterized by a particular vibrational quantum number (indicated in the figure); (c) rotational excitation, also characterized by a particular rotational quantum number. The Figure shows that the rotational levels are more closely spaced and thus transitions

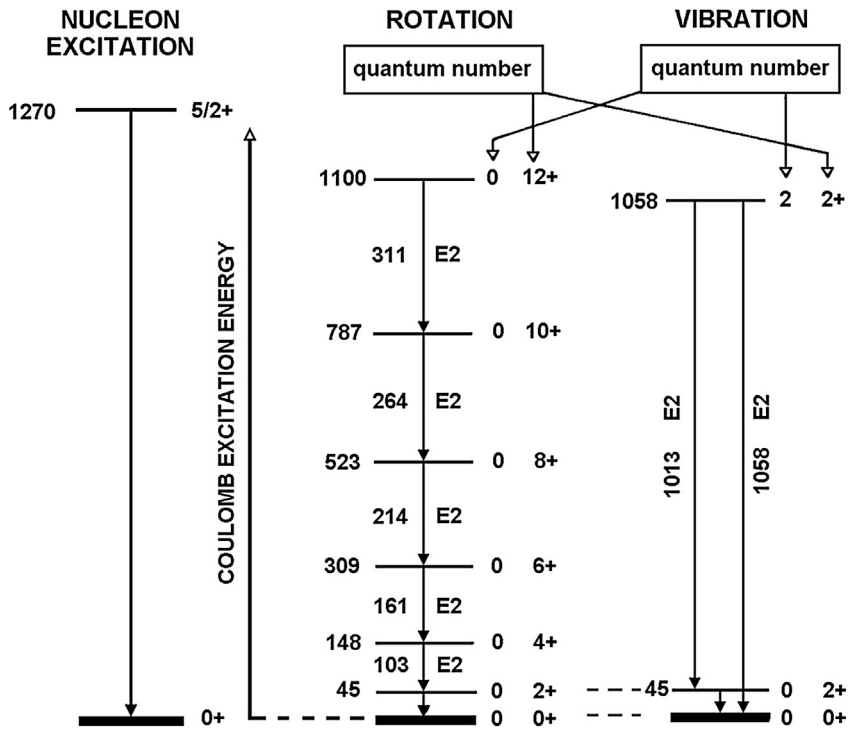


Figure 6.7 Nucleonic, rotational, and vibrational levels observed in ^{238}U; energy in keV. *(From E. K. Hyde.)*

between rotational levels involve lower energies than de-excitation from excited nuclear or vibrational states.

In case of even-even nuclei, the rotational energy levels can be often calculated from the simple expression

$$E_{\text{rot}} = \left(\hbar^2/2I_{\text{rot}}\right) n_{\text{r}}(n_{\text{r}} + 1) \tag{6.41}$$

where I_{rot} is the moment of inertia and n_{r} the rotational quantum number; this equation is identical to (3.39). The validity of this equation depends on whether the different modes of motion can be treated independently or not, which they can for strongly deformed nuclei like ^{238}U.

6.5. THE UNIFIED MODEL OF DEFORMED NUCLEI

The collective model gives a good description for even-even nuclei but cannot account for some of the discrepancy between observed spins and the spin values expected from the single-particle shell model. The latter was developed on the assumption of a nucleon